Collateralised Products

Seminar Module

conducted by

ACF Consultants Ltd





© 2001-23 ACF Consultants Ltd

Topics Covered

- CDO concepts
- Creating a CDO
- Income and principal waterfalls
- Motivation for securitisation
- Synthetic CDOs

© 2001-23 ACF Consultants Ltd

CDO Concepts

© 2001-23 ACF Consultants Ltd

Introducing Collateralised Products

Collateralised Products involving the securitisation of assets whereby:

- A pool of assets are…
- Sold to a Special Purpose Vehicle, which…
- Repackages the assets into tranches having different risk/reward characteristics, and…
- The tranches are bought by investors

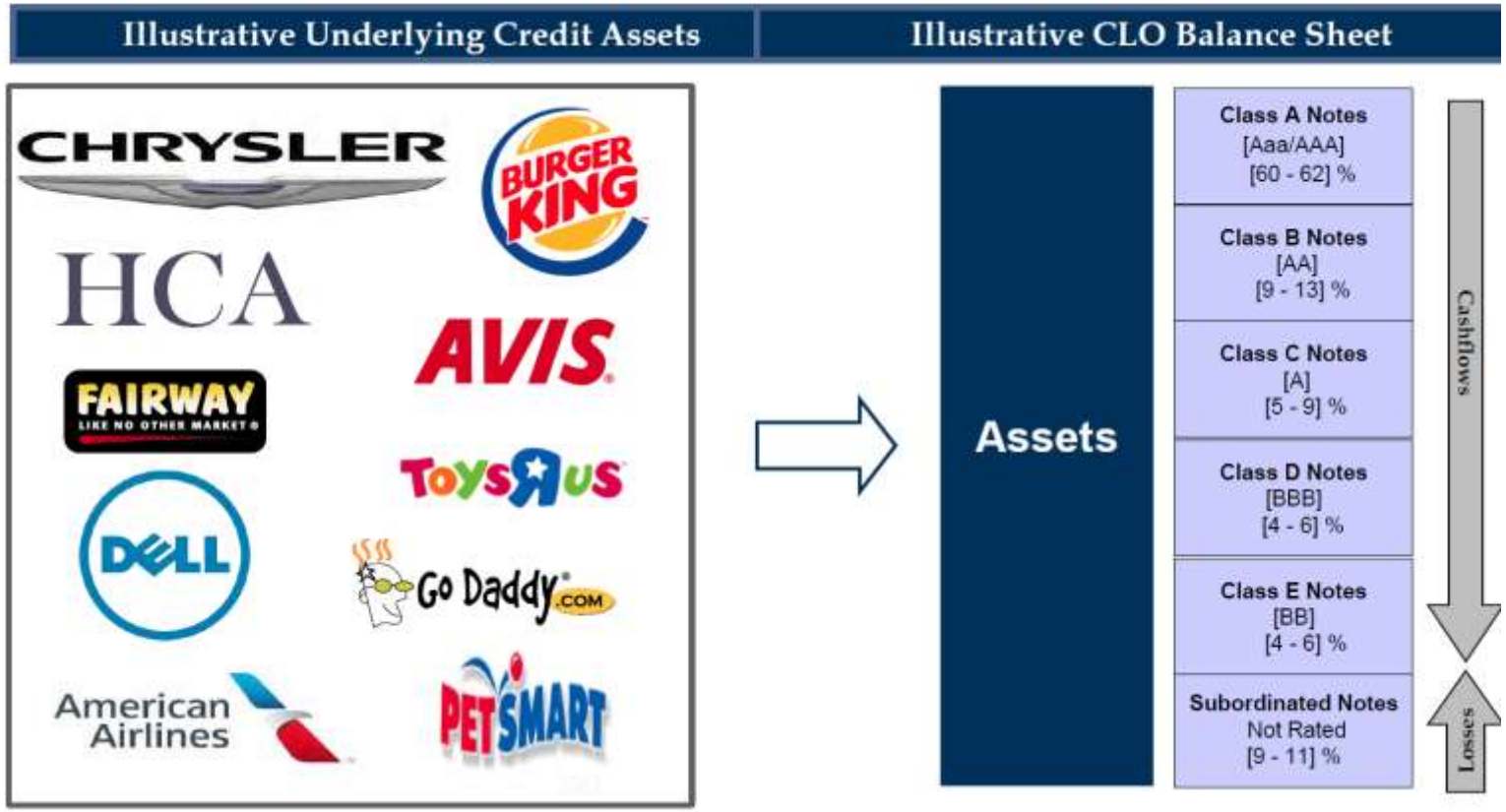
The rationale for investors includes the ability to

- Access return from a wider range of credit markets
- Tailor the risk profile to the investor's requirements
- Maximise returns for that risk profile

© 2001-23 ACF Consultants Ltd

Example – CLOs

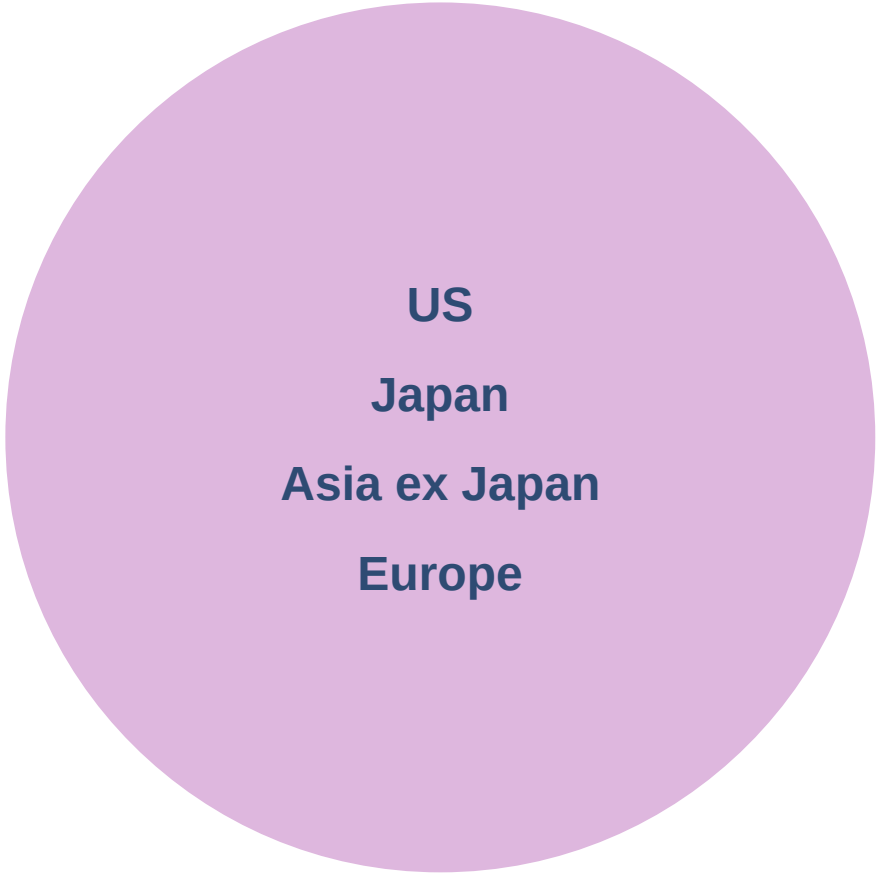
- CLOs invest in pieces of loans to mostly well-known US and international companies
- The interest and principal of the loans pay the interest (and principal) of the CLO notes:
  - Starting at the AAA notes and…
  - Working the way down the waterfall
- If there are losses on loans from defaults, those losses impact the equity holdings first



© 2001-23 ACF Consultants Ltd

CDO Investors

CLO investors come from many regions of the world



Investors include a wide cross-section of entities:



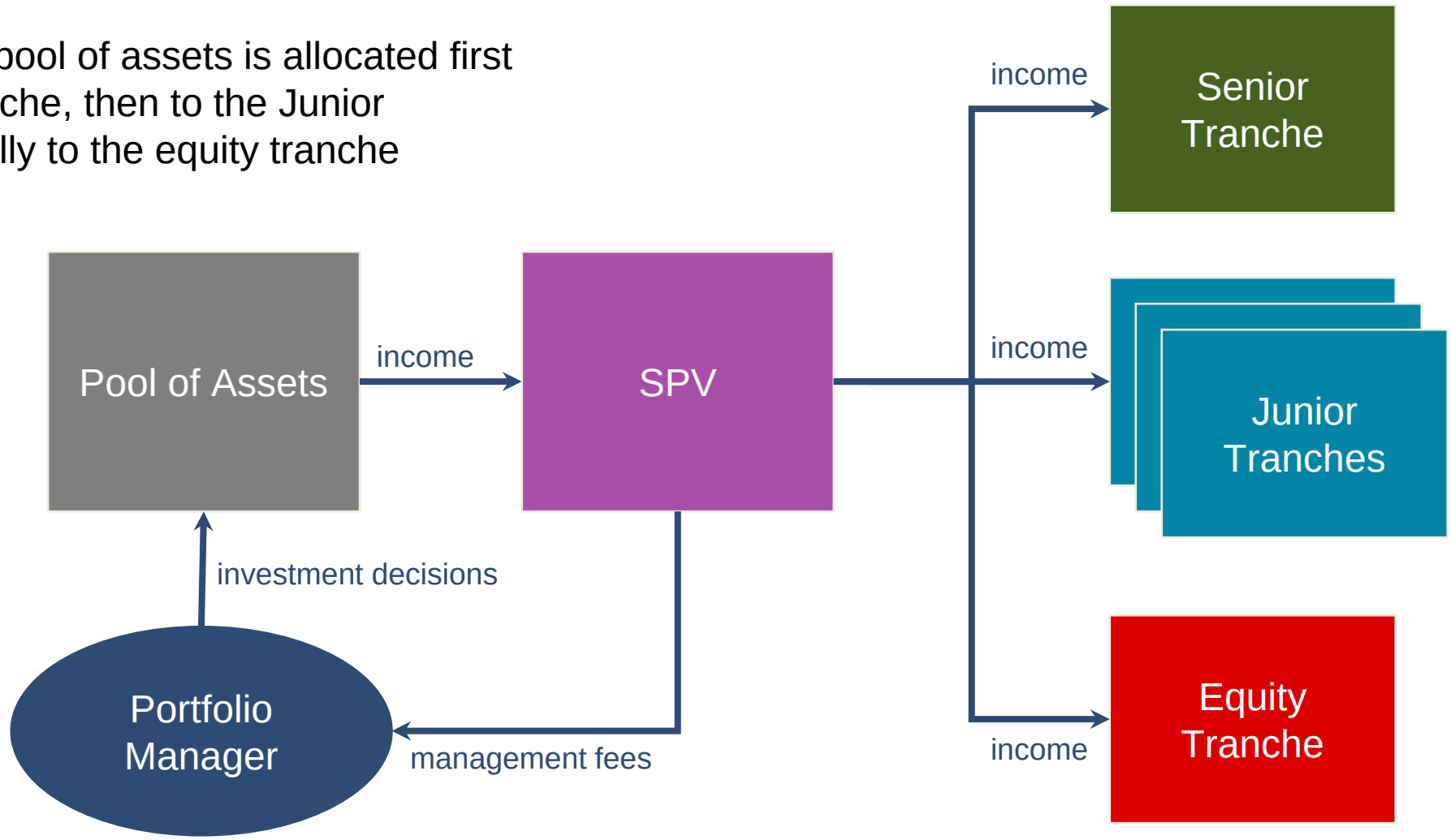
© 2001-23 ACF Consultants Ltd

Creating a CDO

© 2001-23 ACF Consultants Ltd

Creating a CDO

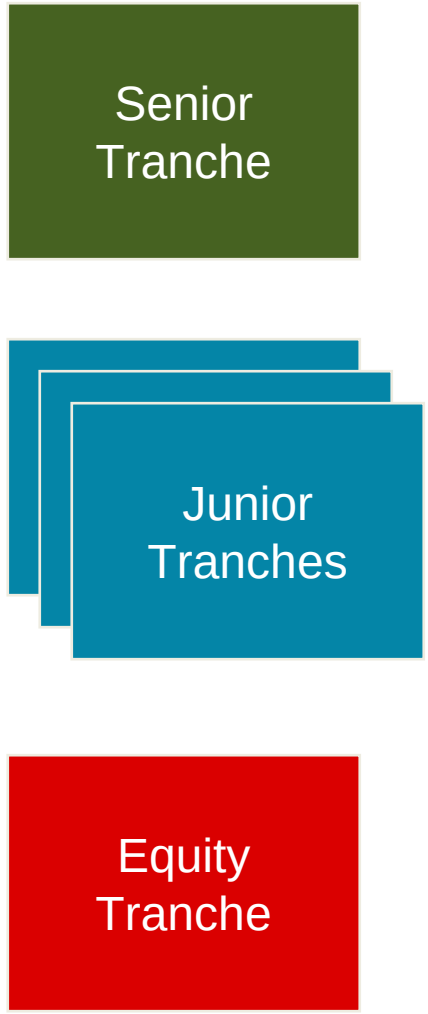
- Income from the pool of assets is allocated first to the Senior tranche, then to the Junior tranches and finally to the equity tranche



© 2001-23 ACF Consultants Ltd

Example CDO Tranches

- **Class A Senior Notes**
  - AAA Rated
  - Libor + 0.40%
  - $80,000,000
- **Class B Secured Notes**
  - A Rated
  - Libor + 0.80%
  - $8,000,000
- **Class C Subordinated Notes**
  - BBB Rated
  - Libor + 2.25%
  - $6,000,000
- **Equity**
  - Not Rated
  - Excess Return
  - $6,000,000



© 2001-23 ACF Consultants Ltd

Example of an Actual CLO Structure

Octagon Investment Partners 43, Ltd.

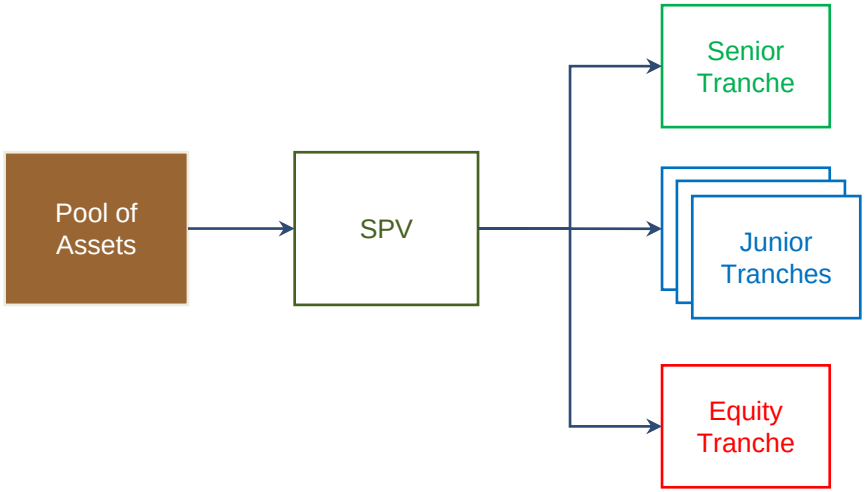
Class	Expected Rating	Expected Rating Outlook	Amount ($ Mil.)	CE (%)[a]	Interest Rate (%)[b]	Final Maturity
A-1	AAAsf	Stable	310.00	38.0	3mL + 1.30	October 2032
A-2	AAAsf	Stable	15.00	35.0	3mL + 1.65	October 2032
B	NRsf	N.A.	55.00	24.0	3mL + 1.75	October 2032
C	NRsf	N.A.	27.25	18.6	3mL + 2.40	October 2032
D	NRsf	N.A.	30.75	12.4	3mL + 3.90	October 2032
E	NRsf	N.A.	22.00	8.0	3mL + 6.60	October 2032
Subordinated	NRsf	N.A	46.05	N.A.	Residual	October 2032
Total			**506.05**			

Source: Fitch Ratings, 26 Aug 2019

© 2001-23 ACF Consultants Ltd

The Waterfall of Cashflows

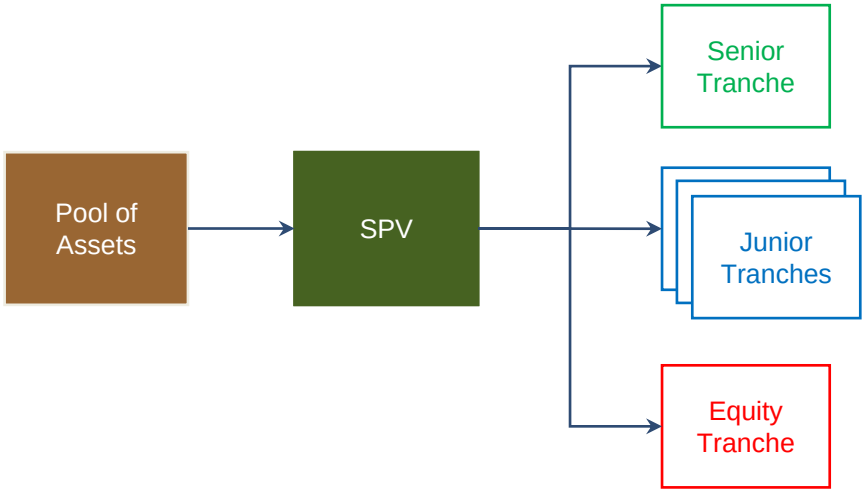
- At the **start** of each period, SPV begins to accrue income from the pool of assets



© 2001-23 ACF Consultants Ltd

The Waterfall of Cashflows (cont'd)

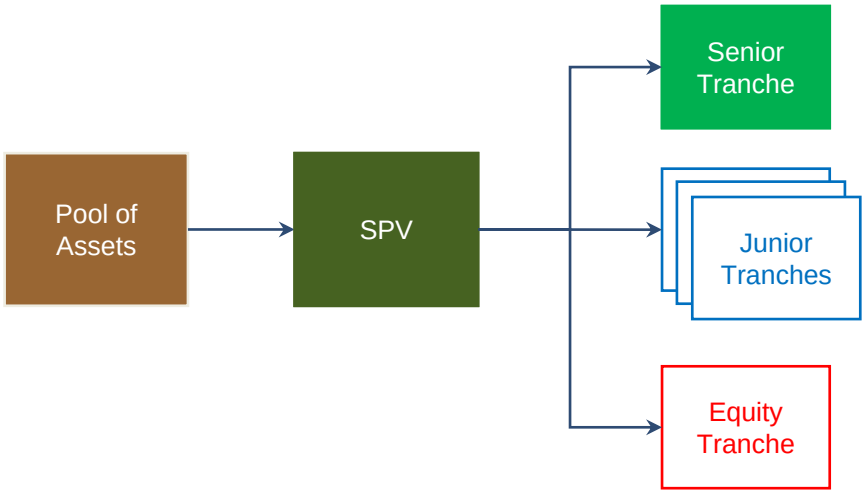
- At the **start** of each period, SPV begins to accrue income from the pool of assets
- By the **end** of each period, SPV accrues the period's income from the asset pool



© 2001-23 ACF Consultants Ltd

The Waterfall of Cashflows (cont'd)

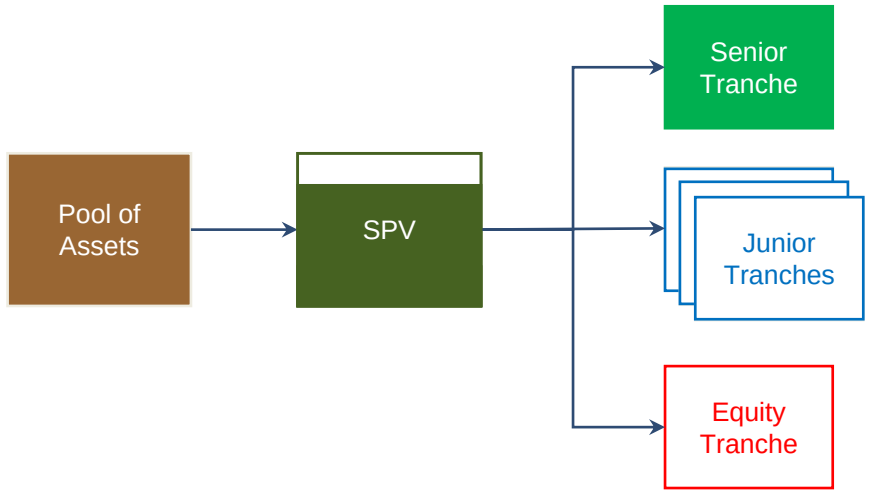
- At the **start** of each period, SPV begins to accrue income from the pool of assets
- By the **end** of each period, SPV accrues the period's income from the asset pool
- Income accrued from the SPV is first paid to the **Senior Tranche**



© 2001-23 ACF Consultants Ltd

The Waterfall of Cashflows (cont'd)

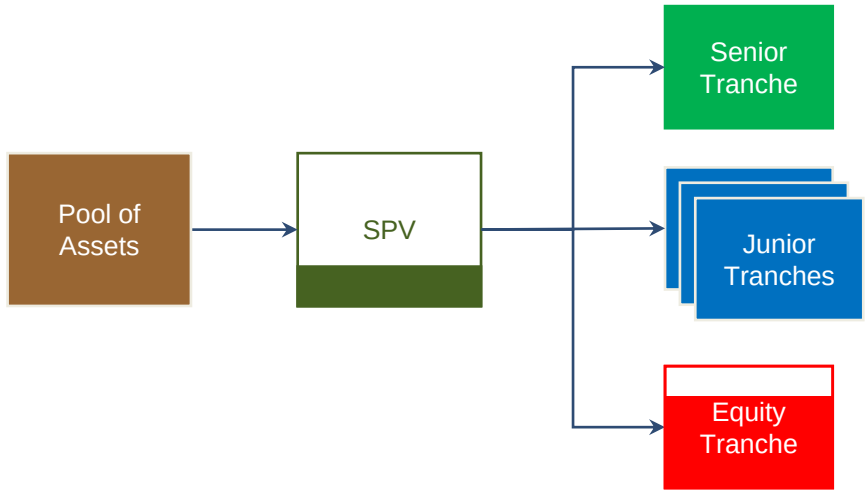
- At the **start** of each period, SPV begins to accrue income from the pool of assets
- By the **end** of each period, SPV accrues the period's income from the asset pool
- Income accrued from the SPV is first paid to the **Senior Tranche**
- Next, income is distributed to the **Junior Tranches**



© 2001-23 ACF Consultants Ltd

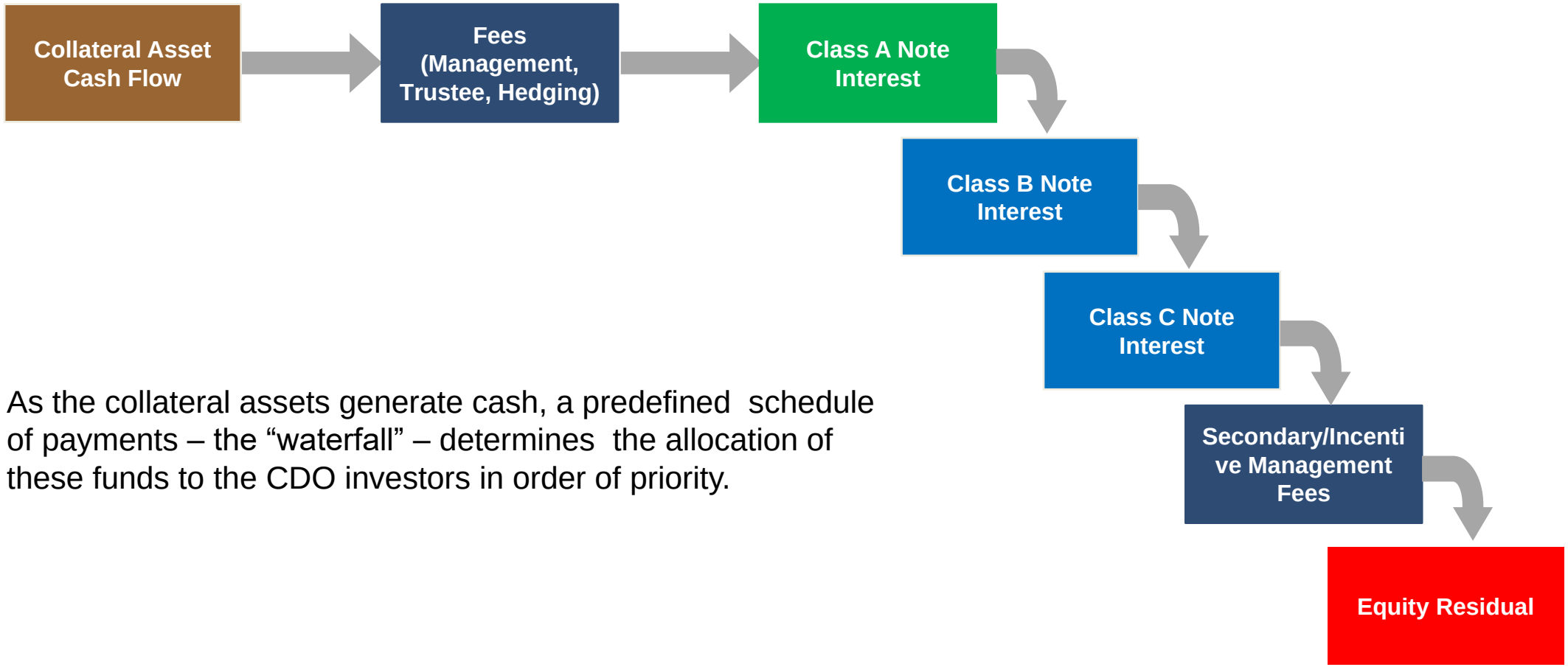
The Waterfall of Cashflows (cont'd)

- At the **start** of each period, SPV begins to accrue income from the pool of assets
- By the **end** of each period, SPV accrues the period's income from the asset pool
- Income accrued from the SPV is first paid to the **Senior Tranche**
- Next, income is distributed to the **Junior Tranches**
- Finally, any residual income from the SPV is paid to the **Equity Tranche**



© 2001-23 ACF Consultants Ltd

The Waterfall of Cashflows (Summary)



As the collateral assets generate cash, a predefined schedule of payments – the “waterfall” – determines the allocation of these funds to the CDO investors in order of priority.

© 2001-23 ACF Consultants Ltd

Coverage Tests / Triggers

- Over Collateralization (OC) test for each class of note
  - Class B test = (Portfolio Principal Balance) / (Outstanding Class A Notes and the Class B Notes)
  - Portfolio Principal Balance must exclude distressed assets
- Interest Coverage (IC) test for each class of note
- Supplemental diversion test (during reinvestment period)
  - Additional minimum OC test for first loss notes
  - Prevents excess spread being distributed too rapidly to the detriment of rated notes

© 2001-23 ACF Consultants Ltd

CDO Tranches

- Different tranches have tiered priority to the stream of income channelled from the asset pool through the SPV, and are structured with:
  - Credit ratings ranging from AAA to unrated
  - Expected earnings from low to high
- Even if the average credit quality of pooled assets is low (e.g. BB)…
- The Senior Tranche can nonetheless earn an AAA rating
  - A significant number of the pooled assets would need to default before investors were impaired
  - Senior Tranche investors are (virtually) guaranteed income and full capital repayment
- The Junior Tranche(s) protects the Senior Tranche
  - Will receive a higher coupon
  - May be some capital impairment if a significant number of credits in the pool default
- The Equity Tranche protects the Junior Tranche
  - It has the highest expected return…
  - …but also suffers the greatest exposure to risk as it is the “first loss” piece
  - Investors can lose their entire investment if the number of defaults in the asset pool is significant

© 2001-23 ACF Consultants Ltd

CDO Tranches

Credit rating on senior tranche depends on degree of collateralization

- AAA-rating requires approximately a 1.50 × collateralization
- AA-rating requires approximately a 1.33× collateralization
- $100m of pooled assets could create:
  - $67m of AAA-rated senior tranche
  - $33m of lower-rated junior tranches
- Or…
  - $75m of AA-rated senior tranche
  - $25m of lower-rated junior tranches
- The bottom tranche(s) are actually pivotal to creating a CBO deal
  - Protects senior tranches
  - Relative difficulty of selling bottom tranche means that these usually have to be pre-placed before structure is launched
- Bottom tranche has to be:
  - Large enough to absorb losses on senior tranches
  - Small enough for issuer to place

© 2001-23 ACF Consultants Ltd

The Special Purpose Vehicle

- A distinct legal entity:
  - Company (in Euromarkets)
  - Trust (in the US) overseen by Trustee (usually agency services department of bank)
  - Not linked to originator
  - Bankruptcy-remote from sponsor
- Acts as the conduit through which securitisation takes place
- Buys the pool of assets
- Transforms the risk/return stream
- Issues the securities which investors buy
- Usually incorporated:
  - Offshore: Jersey, Cayman Islands
  - In SPV-friendly environments: Ireland, Netherlands
    … where withholding tax is low or zero
- Also known as:
  - Special Purpose Entities (SPEs)
  - Special Purpose Companies (SPCs)

© 2001-23 ACF Consultants Ltd

Key Actors

Originator / transferor

- The entity that either generates, or purchases and assembles portfolios of loans or bonds for transfer to or reference for loss determination by the Issuer
- Can remain unnamed in public

Sponsor / arranger

- Structures the deal, may be quite distinct from Originator

Rating agencies

- Moody's, S&P and Fitch: Active players in structuring the transaction who can be involved at an early stage for novel structures

Transferee / issuer

- A bankruptcy remote SPV that issues the notes and avoids taxation at the entity level

Trustees, paying agents etc.

- Administer the ABS

Investment managers

- Invest and manage the proceeds from the structured note sale and prior to closing build collateral through borrowings

Placement agents

- The investment bank or banks that sell the structured notes

Investors

- The ultimate purchasers of the structured notes

© 2001-23 ACF Consultants Ltd

Role of the Investment Manager

- Invest and re-invest according to the structural covenants and waterfall structure
  - Select collateral assets to target portfolio quality or temporarily hold other liquid assets
  - Target surpassing of coverage, diversity and other tests
- Hedge
  - Enter into fixed vs. floating hedges as required by mismatches between the note coupons and collateral
  - Enter into currency swaps as required for foreign currency assets
  - Manage LIBOR vs. LIBOR risk caused by mismatches in payment dates between the note coupons and collateral income
- Discretionary trading
  - Actively manage the portfolio including using derivatives and short positions
  - Respect activity constraints where relevant

© 2001-23 ACF Consultants Ltd

Life-cycle

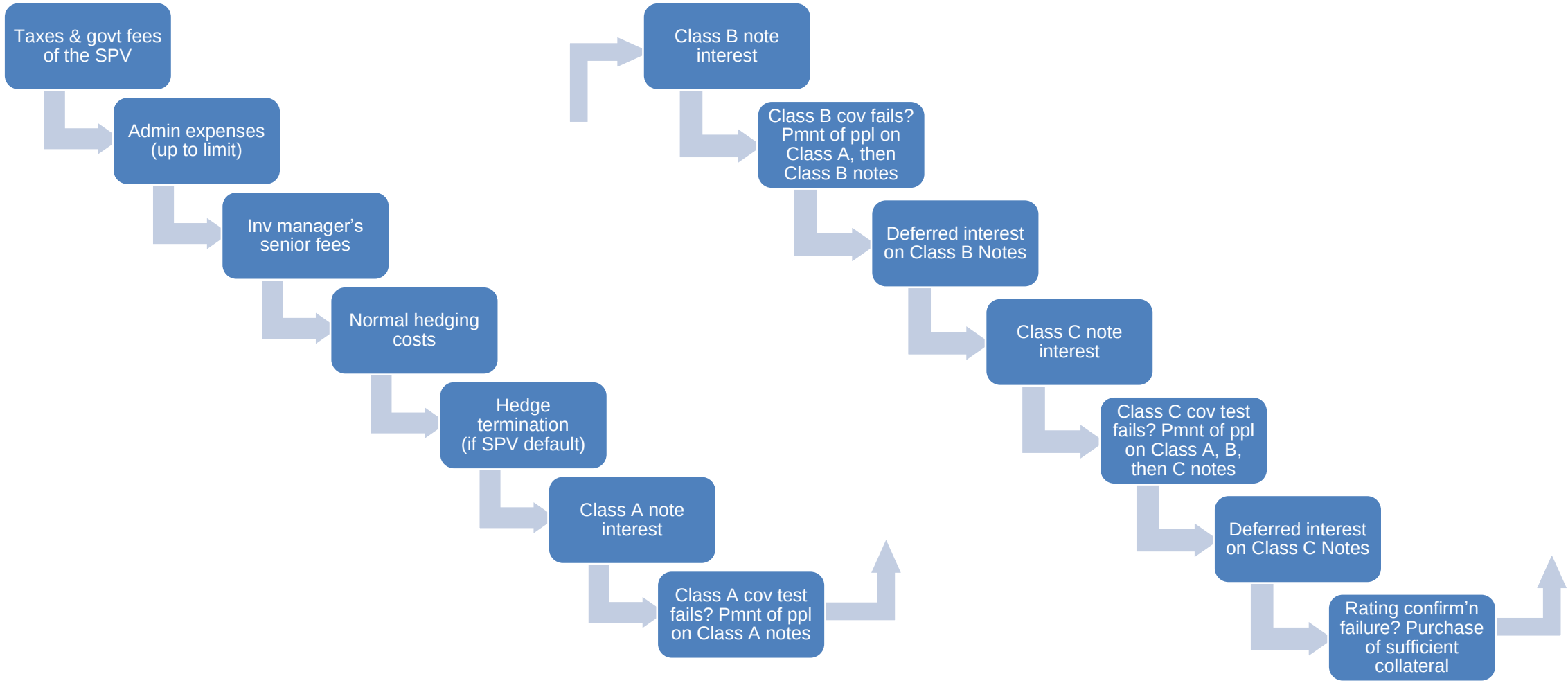
1. Motivation
2. Structuring
   - Select target asset types, and investment manager
   - Structure credit enhancements
   - Legal documents required to set up SPV
3. Pre-sale / Pre-closing
   - Obtain ratings and build order book
   - Warehousing of target assets
4. Closing
   - Allocation to successful bidders and collection of proceeds once assets covering a %age of target achieved
5. Ramp-Up
   - CDO continues to build up its collateral asset pool
   - Ramp-up period can typically take six months
   - Once target portfolio has been acquired, the CDO passes the effective date
6. Non-call period
   - Equity investors can exercise the rights to call a deal after a non-call period (typically 2 years).
   - If called, equity investors repay the debt tranches at par, through refinancing proceeds / sale of the loan portfolio.
7. Reinvestment period
   - The manager actively manages the loan portfolio during the reinvestment period (typically 4+ years), using loan principal repayments and asset sale proceeds to purchase new assets.
   - Early redemption possible due to asset deterioration or other embedded call option
   - Amortisation of principal begins after reinvestment period
8. Maturity / Redemption

© 2001-23 ACF Consultants Ltd

Income and Principal Waterfalls

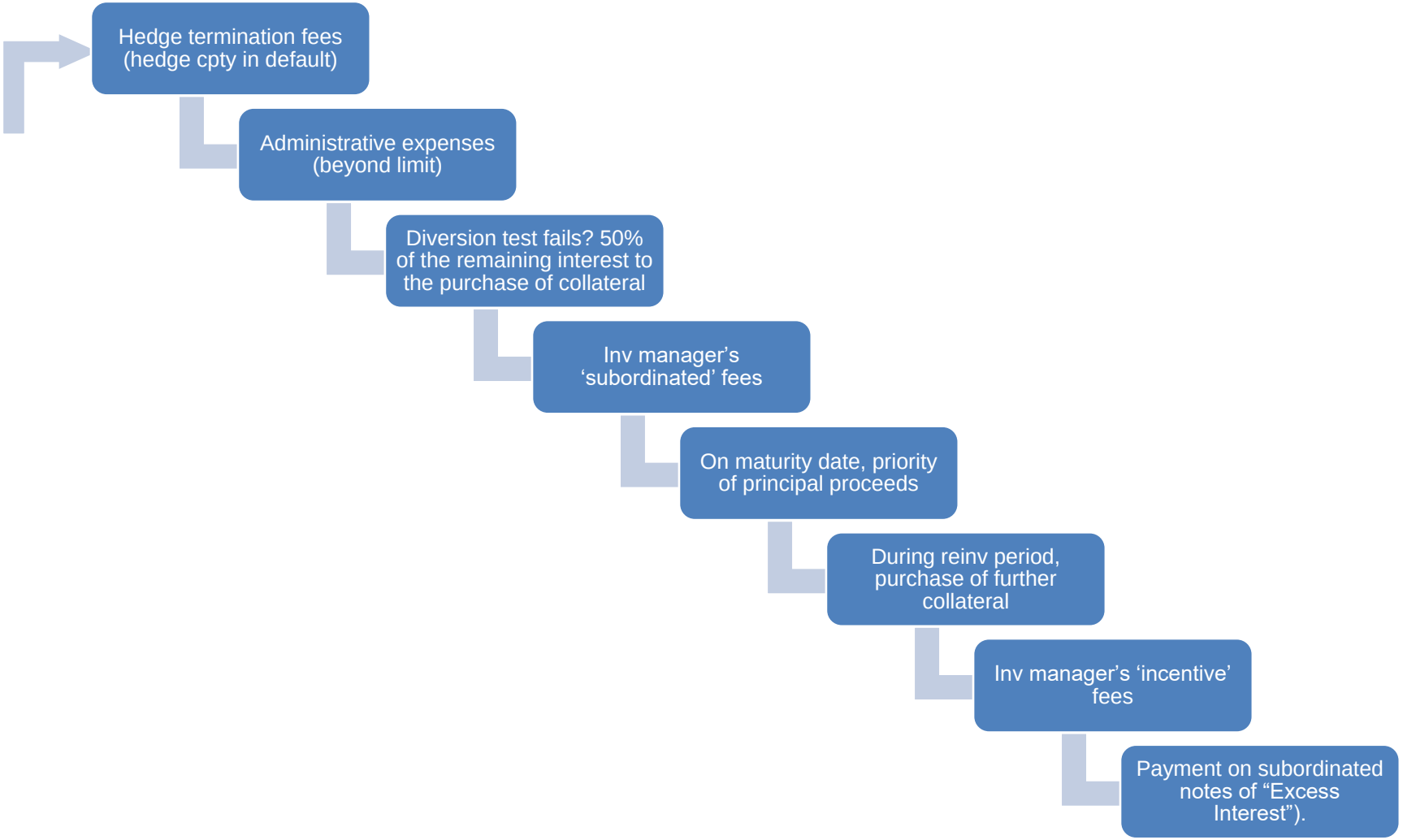
© 2001-23 ACF Consultants Ltd

Income Waterfall



© 2001-23 ACF Consultants Ltd

Income Waterfall (continued)



© 2001-23 ACF Consultants Ltd

Principal Waterfall

1. Items under clauses 1-16, 18-19 (previous page) to the extent not paid from Interest Proceeds…
   - But if an OC test fails, no lower priority payments made, & remaining principal proceeds are distributed to clause 4 or 5 below.
2. Rating confirmation failure on effective date?
   - … to purchase of collateral
3. Payment of redemption price for notes in order of seniority if structure called.
4. During reinvestment period at option of inv manager
   - … to purchase of collateral, or
   - … invested in other eligible investments
5. After the reinvestment period
   - … to payment of principal on the Class A, B and C notes in order of seniority, then
   - … to payment of items under clauses 21 & 24, then
   - … to subordinated notes any remaining principal proceeds.

© 2001-23 ACF Consultants Ltd

Motivation for Securitisation

© 2001-23 ACF Consultants Ltd

Motivation for Securitisation

Principally used by banks for:

- Funding
  - To support rapid asset growth
  - To diversify funding mix
  - To improve or eliminate maturity mismatch
- Balance Sheet Management
  - To shrink balance sheet size
  - To reduce regulatory capital required
  - To improve RoE
- Risk Management
  - Credit risk is transferred to investors

© 2001-23 ACF Consultants Ltd

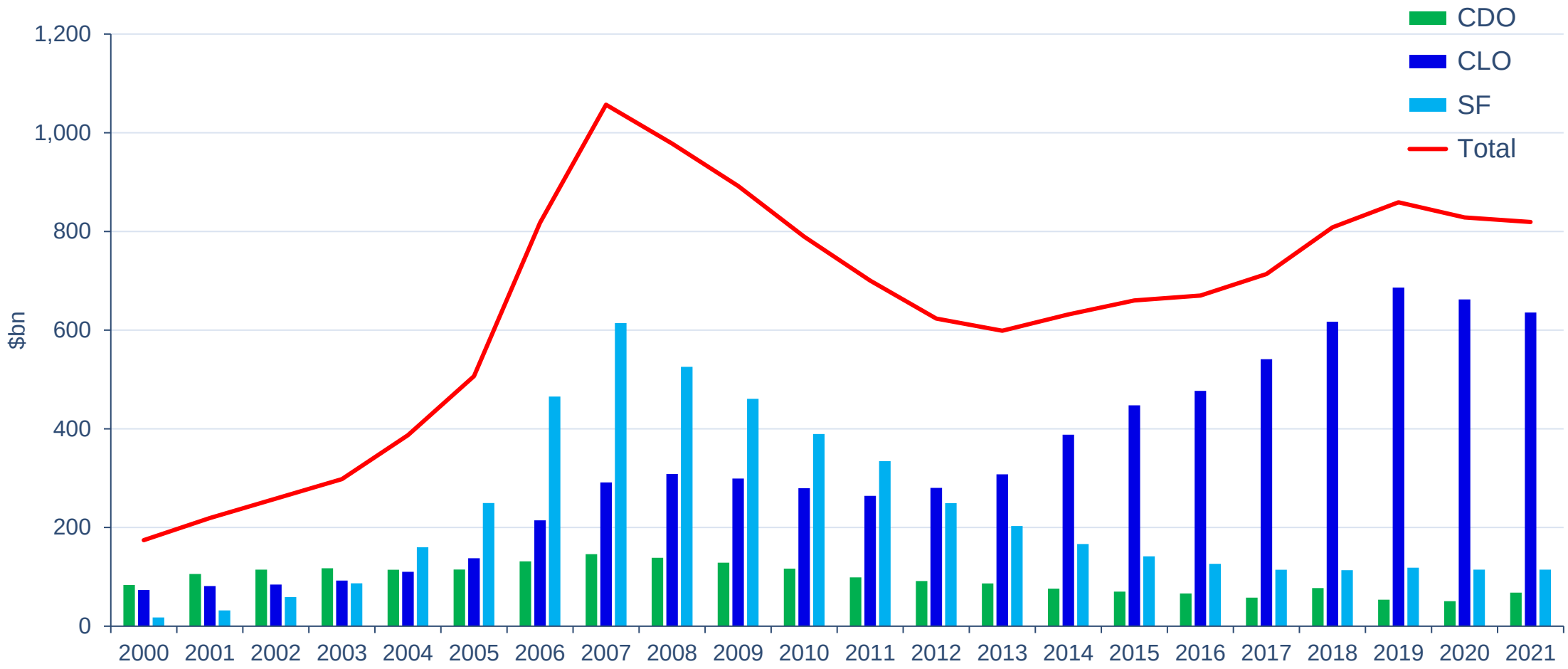
Advantages to Investors of CDOs

Investors gain:

- Relatively high yields on senior notes deliver compared to other ABS of similar credit rating
- Exposure to a diversified range of corporate credit names
- Access to fund management and credit analysis skills of portfolio manager
- A lower level of uncertainty and risk exposure compared to single bonds of a similar credit rating

© 2001-23 ACF Consultants Ltd

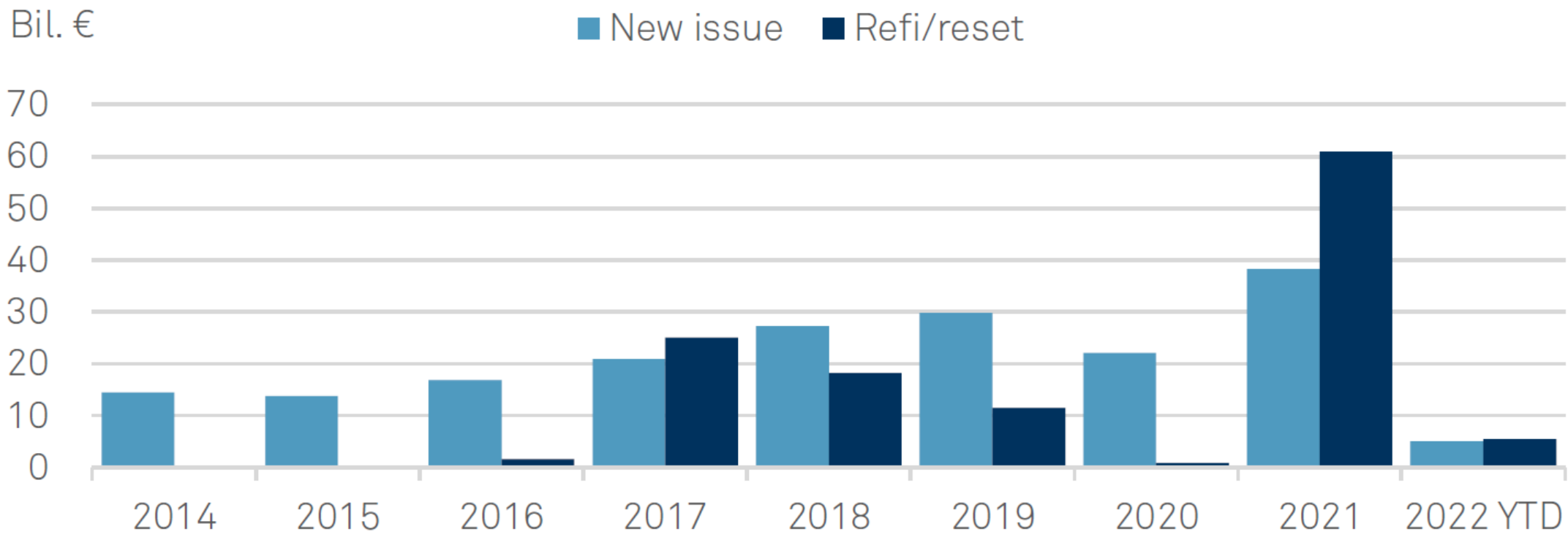
USD CDOs Outstanding



Source: SIFMA

© 2001-23 ACF Consultants Ltd

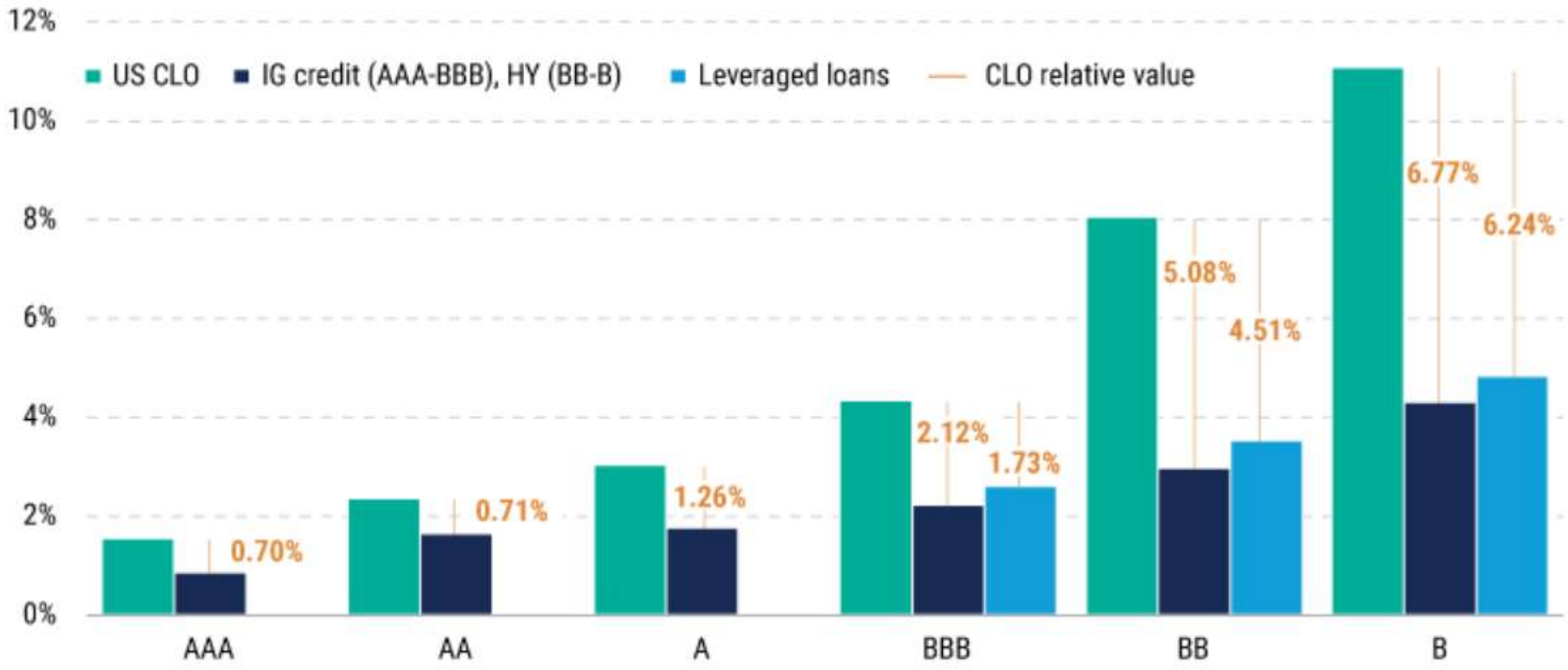
European CLO Issuance



Source: S&P Global Ratings

© 2001-23 ACF Consultants Ltd

Comparing Investment Yields



Source: JP Morgan, Bloomberg, and S&P/LCD as of 31 August 2021

- CLO note yields increase materially as one moves down the rating scale
- At every rating level, CLO notes tend to yield more than the equivalently rated investments in other asset classes

© 2001-23 ACF Consultants Ltd

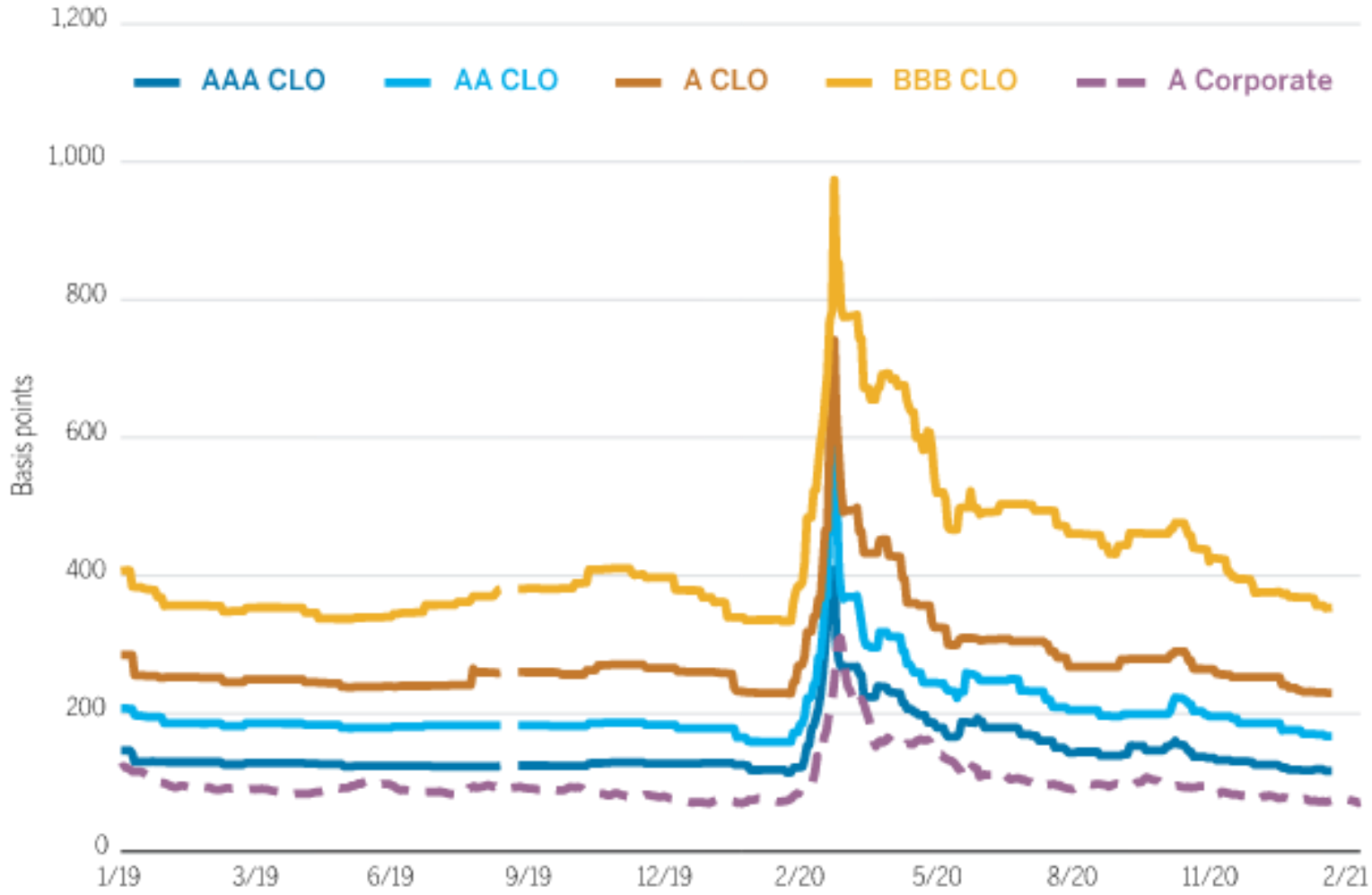
CLOs vs. other Credit Asset Classes

	Senior secured loans	High yield bonds and distressed debt	Emerging market debt	CLO tranches
Issuer	Syndicated loans issued primarily by companies in the US and Western Europe	Corporate bonds issued primarily in the developed markets (US & Europe)	Corporate bonds issued primarily in Latin America, Europe, Asia, and the Middle East and Africa	Securitization trust (loans issued primarily by companies in the US and Western Europe serve as collateral)
Ranking	Typically senior secured	1st & 2nd loss	1st loss	1st, 2nd, 3rd, 4th, 5th, 6th & 7th loss
Maturity	Typically issued with a 5- to 9-year maturity	5-10 years	5-10 years	Typically issued with a legal maturity of 12-13 years; weighted average life of 6-10 years
Interest rate duration	0.25 years	0-6 years	0-6 years	0.25 years
Coupon type	Floating rate	Fixed rate	Fixed rate	Floating rate
Spread/OAS	Libor + 417 bps	264 bps	249 bps	• Libor + 103 bps (AAA) • Libor + 151 bps (AA) • Libor + 187 bps (A) • Libor + 293 bps (BBB) • Libor + 620 bps (BB) • Libor + 905 bps (B)
Yield to worst	4.23%	4.23%	3.63%	• 1.72% (AAA) • 2.59% (AA) • 3.26% (A) • 4.48% (BBB) • 8.23% (BB) • 11.30% (B) • 12-16% (Equity)
Credit rating	BB-CCC	BB-CCC	BBB-B	AAA-B; Not Rated for Equity
Historical default rate	~3% p.a.	~3% p.a.	~2.5%*	0.3%**
Historical recovery rate	~75%	~35%	~35%	23%**
Additional notes	Limited call protection. Extended trade settlement times (T+7)	Higher price volatility	Higher price volatility	Can be less liquid when compared to underlying bank loans (T+2)

Source: Pinebridge

© 2001-23 ACF Consultants Ltd

CLO Spreads



Sources: JPMorgan CLO Index (CLOIE), Bloomberg Barclays A Rated Corporate Index | Chart data: January 2019 – February 2021

© 2001-23 ACF Consultants Ltd

Synthetic CDOs

© 2001-23 ACF Consultants Ltd

Synthetic CDOs

In a **Synthetic CDO**:

- Credit risk from a pool of assets is transferred to investors
- Assets themselves are **not** transferred
- Synthetic CDOs use credit derivatives to transfer credit risk while leaving ownership of the asset pool with the originator

Synthetic CDOs:

- First executed in the US in 1997 (JPMorgan's BISTRO deal)
- First appeared in Europe in 1998
- Are particularly attractive in Europe because:
  - Legal jurisdiction makes true-sale cash-flow securitisation more difficult
  - US commercial banks are typically rated lower than their European counterparts – so funding element of cash-flow securitisation attractive
  - European commercial banks generally have higher ratings – so less need for funding component of cash-flow transaction

© 2001-23 ACF Consultants Ltd

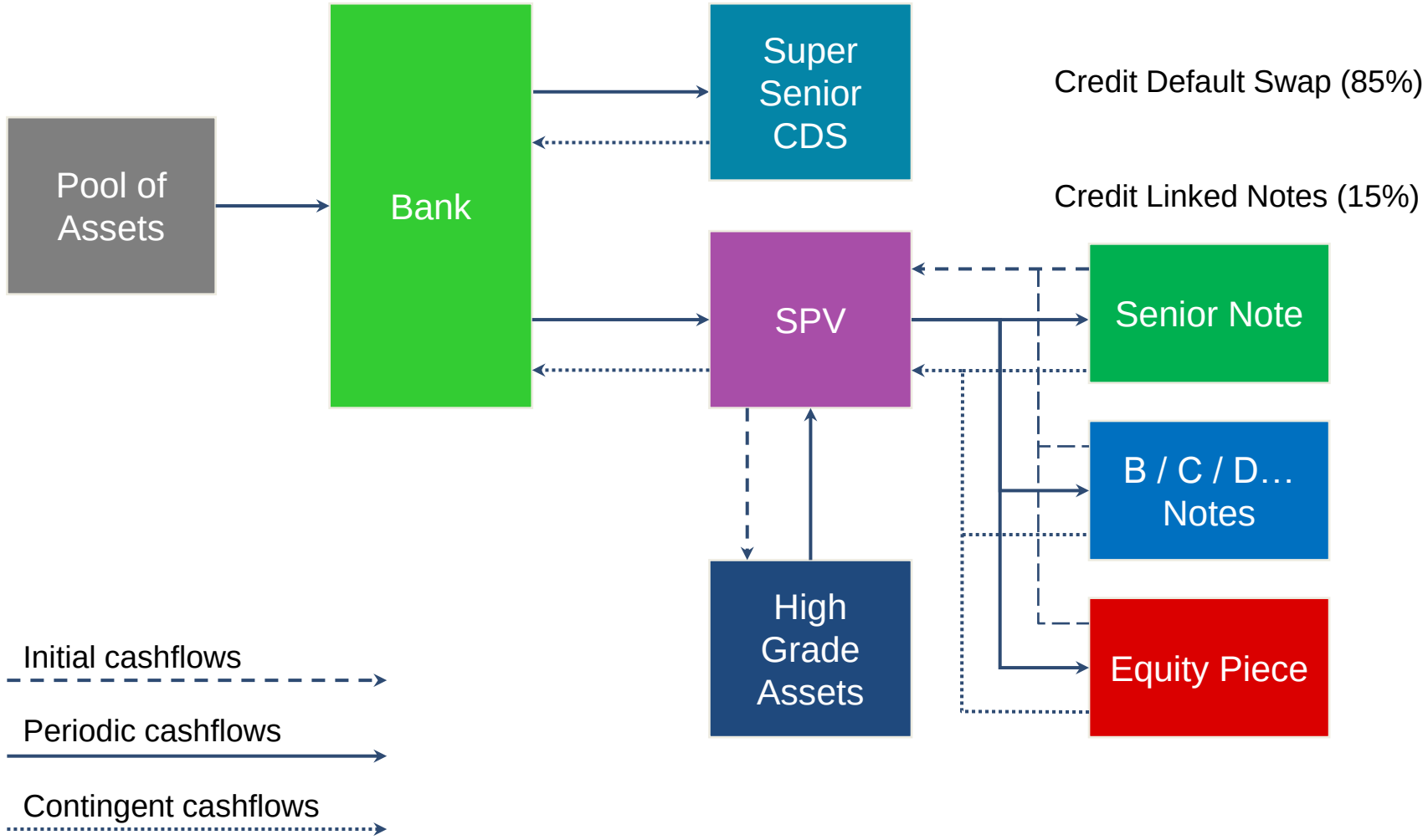
Types of Synthetic CDO

Synthetic CDOs can be:

- Unfunded
  - CDS on 100% of the asset pool are used to transfer the credit risk
- Partially funded
  - CDS purchased on part of the asset pool
  - Notes issued against the balance (as in cash-flow CDO)
- Fully funded
  - Credit risk transferred through issue of Credit Linked Notes (CLNs)

© 2001-23 ACF Consultants Ltd

Creating a Synthetic CDO



© 2001-23 ACF Consultants Ltd

Creating a Synthetic CDO (cont'd)

- Bank purchases a CDS on the major part of the asset pool
  - As CDS is "super senior" it carries very low swap premium
  - Example:
    - AAA premium: 45-50 bp
    - Super senior premium: 10-12 bp
- SPV issues CLNs on the remainder of the asset pool
- Proceeds from note issue invested in high-grade assets to de-link rating of CLN from that of issuer
- If no default occurs:
  - Note investors receive enhanced return on CLN provided by:
    - Return from high-grade assets
    - Swap premiums from bank
- If defaults occur:
  - Bank pays first loss (owns equity piece)
  - CLNs provide protection on further losses
  - Super senior CDS provides backstop credit protection for the bank

© 2001-23 ACF Consultants Ltd

Advantages of Synthetic CDOs

- Speed of implementation
  - Cashflow securitisations can take 3-4 months to complete
  - Synthetic CDOs can be implemented in 6-8 weeks
- Client relationships maintained
  - … as assets remain on the balance sheet of originator
- No (or reduced) requirement to fund
- Transaction costs reduced
  - … in fully unfunded cases there is no need to establish an SPV
- A wider range of reference assets can be covered than in true-sale securitisation
  - Loans
  - Undrawn lines of credit
  - Bank guarantees
  - etc.
- Overall cost of buying credit protection is lowered
- Still provides regulatory capital reduction benefits

© 2001-23 ACF Consultants Ltd